

Tumor development is associated with decrease of TET gene expression and 5-methylcytosine hydroxylation.

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The TET (ten-eleven translocation) family of α -ketoglutarate (α -KG)-dependent dioxygenases catalyzes the sequential oxidation of 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine and 5-carboxylcytosine, leading to eventual DNA demethylation. The TET2 gene is a bona fide tumor suppressor frequently mutated in leukemia, and TET enzyme activity is inhibited in IDH1/2-mutated tumors by the oncometabolite 2-hydroxyglutarate, an antagonist of α -KG, linking 5mC oxidation to cancer development. We report here that the levels of 5hmC are dramatically reduced in human breast, liver, lung, pancreatic and prostate cancers when compared with the matched surrounding normal tissues. Associated with the 5hmC decrease is the substantial reduction of the expression of all three TET genes, revealing a possible mechanism for the reduced 5hmC in cancer cells. The decrease of 5hmC was also observed during tumor development in different genetically engineered mouse models. Together, our results identify 5hmC as a biomarker whose decrease is broadly and tightly associated with tumor development.

Mechanisms and functions of Tet protein-mediated 5-methylcytosine oxidation.

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Ten-eleven translocation 1-3 (Tet1-3) proteins have recently been discovered in mammalian cells to be members of a family of DNA hydroxylases that possess enzymatic activity toward the methyl mark on the 5-position of cytosine (5-methylcytosine [5mC]), a well-characterized epigenetic modification that has essential roles in regulating gene expression and maintaining cellular identity. Tet proteins can convert 5mC into 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and 5-carboxylcytosine (5caC) through three consecutive oxidation reactions. These modified bases may represent new epigenetic states in genomic DNA or intermediates in the process of DNA demethylation. Emerging biochemical, genetic, and functional evidence suggests that Tet proteins are crucial for diverse biological processes, including zygotic epigenetic reprogramming, pluripotent stem cell differentiation, hematopoiesis, and development of leukemia. Insights into how Tet proteins contribute to dynamic changes in DNA methylation and gene expression will greatly enhance our understanding of epigenetic regulation of normal development and human diseases.

TET enzymes regulate skeletal development through increasing chromatin accessibility of RUNX2 target genes.

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The Ten-eleven translocation (TET) family of dioxygenases mediate cytosine demethylation by catalyzing the oxidation of 5-methylcytosine (5mC). TET-mediated DNA demethylation controls the proper differentiation of embryonic stem cells and TET members display functional redundancy during early gastrulation. However, it is unclear if TET proteins have functional significance in mammalian skeletal development. Here, we report that Tet genes deficiency in mesoderm mesenchymal stem cells results in severe defects of bone development. The existence of any single Tet gene allele can support early bone

formation, suggesting a functional redundancy of TET proteins. Integrative analyses of RNA-seq, Whole Genome Bisulfite Sequencing (WGBS), 5hmC-Seal and Assay for Transposase-Accessible Chromatin (ATAC-seq) demonstrate that TET-mediated demethylation increases the chromatin accessibility of target genes by RUNX2 and facilitates RUNX2-regulated transcription. In addition, TET proteins interact with RUNX2 through their catalytic domain to regulate cytosine methylation around RUNX2 binding region. The catalytic domain is indispensable for TET enzymes to regulate RUNX2 transcription activity on its target genes and to regulate bone development. These results demonstrate that TET enzymes function to regulate RUNX2 activity and maintain skeletal homeostasis.

Development of a rapid mass spectrometric method for the analysis of ten-eleven translocation enzymes.

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In DNA, methylation at the fifth position of cytosine (5mC) by DNA methyltransferases is essential for eukaryotic gene regulation. Methylation patterns are dynamically controlled by epigenetic machinery. Erasure of 5mC by Fe²⁺ and 2-ketoglutarate (2KG) dependent dioxygenases in the ten-eleven translocation family (TET1-3), plays a key role in nuclear processes. Through the event of active demethylation, TET proteins iteratively oxidize 5mC to 5-hydroxymethyl cytosine (5hmC), 5-formylcytosine (5fC) and 5-carboxycytosine (5caC), each of which has been implicated in numerous diseases when aberrantly generated. A wide range of biochemical assays have been developed to characterize TET activity, many of which require multi-step processing to detect and quantify the 5mC oxidized products. Herein, we describe the development and optimization of a sensitive MALDI mass spectrometry-based technique that directly measures TET activity and eliminates tedious processing steps. Employing optimized assay conditions, we report the steady-state activity of wild type TET2 enzymes to furnish 5hmC, 5fC and 5caC. We next determine IC₅₀ values of several small-molecule inhibitors of TETs. The utility of this assay is further demonstrated by analyzing the activity of V1395A which is an activating mutant of TET2 that primarily generates 5caC. Lastly, we describe the development of a secondary assay that utilizes bisulfite chemistry to further examine the activity of wildtype TET2 and V1395A in a base-resolution manner. The combined results demonstrate that the activity of TET proteins can be gauged, and their products accurately quantified using our methods.

Developmental DNA demethylation is a determinant of neural stem cell identity and gliogenic competence.

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DNA methylation is extensively reconfigured during development, but the functional significance and cell type-specific dependencies of DNA demethylation in lineage specification remain poorly understood. Here, we demonstrate that developmental DNA demethylation, driven by ten-eleven translocation 1/2/3 (TET1/2/3) enzymes, is essential for establishment of neural stem cell (NSC) identity and gliogenic potential. We find that loss of all three TETs during NSC specification is dispensable for neural induction and neuronal differentiation but critical for astrocyte and oligodendrocyte formation, demonstrating a selective loss of glial competence. Mechanistically, TET-mediated demethylation was essential for commissioning neural-specific enhancers in proximity to master neurodevelopmental and glial

transcription factor genes and for induction of these genes. Consistently, loss of all three TETs in embryonic NSCs in mice compromised glial gene expression and corticogenesis. Thus, TET-dependent developmental demethylation is an essential regulatory mechanism for neural enhancer commissioning during NSC specification and is a cell-intrinsic determinant of NSC identity and gliogenic potential.

[Role of 5-hydroxymethylcytosine and TET proteins in epigenetic regulation of gene expression].

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DNA methylation plays an important role in epigenetic regulation of human gene expression. Mechanism of active demethylation of the human genome have been a matter of discussion for many years. Recently, a novel group of TET protein family enzymatically converting 5-methylcytosine into 5-hydroxymethylcytosine was discovered, playing a role in active DNA demethylation pathway. Results obtained in subsequent studies pointed that 5-hydroxymethylcytosine was not only an intermediate in that pathway, but might also modify epigenetic profile of the human genome.

Enzymatic analysis of Tet proteins: key enzymes in the metabolism of DNA methylation.

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One of the most exciting recent advances in the epigenetic field is the discovery that 5-methylcytosine (5mC) in DNA can be iteratively oxidized by a family of proteins known as Tet proteins to generate 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and 5-carboxylcytosine (5caC). These 5mC derivatives can be further processed by thymine-DNA glycosylase (TDG) followed by base excision repair or by replication-dependent dilution leading to DNA demethylation. Given the similarity between 5mC and its oxidation derivatives, many of the conventional techniques used for 5mC analysis cannot distinguish between 5mC and 5hmC/5fC/5caC. Here, we describe 2D-TLC and mass spectrometry methods that we have successfully used in differentiating 5mC from its oxidative derivatives as well as in characterizing the enzymatic activity of Tet proteins both in vitro and in vivo.

TET Methylcytosine Oxidases in T Cell and B Cell Development and Function.

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DNA methylation is established by DNA methyltransferases and is a key epigenetic mark. Ten-eleven translocation (TET) proteins are enzymes that oxidize 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC) and further oxidation products (oxi-mCs), which indirectly promote DNA demethylation. Here, we provide an overview of the effect of TET proteins and altered DNA modification status in T and B cell development and function. We summarize current advances in our understanding of the role of TET proteins and 5hmC in T and B cells in both physiological and pathological contexts. We describe how TET proteins and 5hmC regulate DNA modification, chromatin accessibility, gene expression, and transcriptional networks and discuss potential underlying mechanisms and open questions in the field.

Regulation of TET protein stability by calpains.

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DNA methylation at the fifth position of cytosine (5mC) is an important epigenetic modification that affects chromatin structure and gene expression. Recent studies have established a critical function of the Ten-eleven translocation (Tet) family of proteins in regulating DNA methylation dynamics. Three Tet genes have been identified in mammals, and they all encode for proteins capable of oxidizing 5mC as part of the DNA demethylation process. Although regulation of Tet expression at the transcriptional level is well documented, how TET proteins are regulated at posttranslational level is poorly understood. In this study, we report that all three TET proteins are direct substrates of calpains, a family of calcium-dependent proteases. Specifically, calpain1 mediates TET1 and TET2 turnover in mouse ESCs, and calpain2 regulates TET3 level during differentiation. This study provides evidence that TET proteins are subject to calpain-mediated degradation.

TET (Ten-eleven translocation) family proteins: structure, biological functions and applications.

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Ten-eleven translocation (TET) family proteins (TETs), specifically, TET1, TET2 and TET3, can modify DNA by oxidizing 5-methylcytosine (5mC) iteratively to yield 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and 5-carboxycytosine (5caC), and then two of these intermediates (5fC and 5caC) can be excised and return to unmethylated cytosines by thymine-DNA glycosylase (TDG)-mediated base excision repair. Because DNA methylation and demethylation play an important role in numerous biological processes, including zygote formation, embryogenesis, spatial learning and immune homeostasis, the regulation of TETs functions is complicated, and dysregulation of their functions is implicated in many diseases such as myeloid malignancies. In addition, recent studies have demonstrated that TET2 is able to catalyze the hydroxymethylation of RNA to perform post-transcriptional regulation. Notably, catalytic-independent functions of TETs in certain biological contexts have been identified, further highlighting their multifunctional roles. Interestingly, by reactivating the expression of selected target genes, accumulated evidences support the potential therapeutic use of TETs-based DNA methylation editing tools in disorders associated with epigenetic silencing. In this review, we summarize recent key findings in TETs functions, activity regulators at various levels, technological advances in the detection of 5hmC, the main TETs oxidative product, and TETs emerging applications in epigenetic editing. Furthermore, we discuss existing challenges and future directions in this field.
